LUT
University

CT60A5401 GAME DEVELOPMENT PROJECT

ACADEMIC YEAR 2022-2023

Assoc. Prof. Jussi Kasurinen (Adj. prof. of entertainment software engineering)

TODAY'S AGENDA

GAME DEVELOPMENT TOOLS

GAME DESIGN PRINCIPLES

DEVELOPMENT TOOLS

SOFTWARE & GAME DEVELOPMENT

As mentioned in the first lecture, game development and software development share some common features. In fact there are more things in common than many people in the either industry willingly admits.

- **Design work**
- **Development, programming tasks**
  - Continuous delivery as a process model!
- **Tools**
- **Testing**
- **Objectives:**
  - In time
  - In budget
  - With the intended features
  - With acceptable quality

SOFTWARE DEVELOPMENT

Obviously there also are some differences.

- **More emphasis on customers, market**
  - No readymade designs, or “mandatory decisions”
- **Design with artistic vision on how things happen.**
  - “Making dreams into reality”
- **Tools not necessarily the same as in software dev.**
- **Testing with users.**
- **Sound design.**

SOFTWARE DEVELOPMENT

- **Development has certain aspects which are unusual:**
  - Real time client-to-client multiplayer systems
  - Artificial intelligence
  - 3D mathematics
- **Objectives:**
  - In time, in budget, acceptable quality, all features, and.
  - Is the game fun?

WHAT DO YOU NEED TO MAKE GAMES?

In short: Computer, developer and a place to sit

In more detail:

- Game designers
- Graphics, animation artists
- Programmers
- Test audiences
- Programming tools (version control, code editors)
- Game engine or other framework
- Auxiliary services such as physics libraries, payment management systems, cloud services etc.

COMMON ROLES AND TOOLS

Game studios look for focused professionals, startups "all around experts" on

- Project, art directors
- Developers
- Artists
- Musicians, sound effect & editing
- People to run the business aspects

Some common tools

- Drawing tools (Photoshop, 3D Studio), engines (Unity, Unreal), Sound editing (Audacity), programming languages (Java, C#...)

EXAMPLE

DEVELOPER ECOSYSTEMS IN 7 GAME STUDIOS

INTERVIEWS, DATA COLLECTION

	Interviewee role	Description	Main themes of the interviews
1	Team leader or project manager	The interviewee is responsible for the management of the development of one product, or one phase of develpment for all products.	Development process, test process, quality, outsourcing, development tools, organizational aspects.
2	Developer or tester	The interviewee was responsible for the development tasks, preferably also with the responsibilities of software testing activities.	Development process, test process, development tools, development methods, quality.
3	Upper management or owner	The interviewee was from the upper management, or a business owner with an active role in the organization.	Organization, quality, marketing, innovation and design process, development process.
4	Lead designer or Art designer	The interviewee was a game designer, or managerial level person with the ability to affect the product design.	Development process, design and innovation, testing, quality

INTERVIEWS, DATA COLLECTION

Case	Release platforms	Production team size	Maturity, amount of released games
Case A	PC, consoles	Large	Established, more than 10 released products
Case B	Mobile platforms	Small	Recent startup, Less than 5 released products
Case C	Consoles, PC	Large	Established, Less than 10 released products.
Case D	Mobile platforms, PC	Medium	Startup, developing first product
Case E	Mobile platforms	Small	Recent startup, less than 5 released products
Case F	PC	Medium	Startup, developing first product
Case G	Browser games	Small	Startup, developing first product

	Source of the game engine	Tool selection principles	Tool-related development needs	Applied design methods	Most used knowledge transfer methods
Case A	3rd party, locally modified	Compatibility with other tools, popularity.	Availability of the extension modules for the engine	Functional prototypes	Project management tool
Case B	Was own, now 3rd party	Price, portability to different platforms	Amount of bugs, better version control	Design document, screen mock-ups	Daily meetings, messaging system
Case C	3rd party, locally modified	Compatibility with other tools, personal preference	Tools should be more user-friendly and intuitive	Functional prototypes	Project management tool
Case D	3rd party	Versatility, portability	Amount of bugs, restrictive editor	Screen mock-ups, prototypes	Daily meetings, design documents
Case E	Own	Personal preference	Need to adopt a new game engine	Design documents	Email, messaging systems
Case F	3rd party, locally modified	Personal preference, compatibility with other tools	None	Design documents, brainstroming	Messaging system, design documents
Case G	Own	Personal preference	None	Design documents, brainstroming	Email, messaging systems

TOOLS IN DIFFERENT STAGES OF DEVELOPMENT

Activity	Preproduction, design phase	Production, implementation phase
Design, prototyping tools	Documentation tools, Doc. distribution, graphics software, development tools, game engine	Development tools, game engine, graphics software
Implementation, development tools	Graphics software, development tools, game engine, cloud services	Game engine, development tools, graphics software, audio software
Project support tools	Documentation, cloud services, file sharing service	Version control, cloud services, file sharing service, ticket systems

OBSERVATIONS

- Game engine and other complex systems are usually based on third-party solutions as they are considered difficult and expensive to develop independently.
  - There are completely valid jobs on “tool users” in games industry.
- The tool selection principles are not based only on the applied methods or process requirements. → Artists use what they are most comfortable with.
- Some form of prototyping is commonly applied and it is important that the technical infrastructure supports fast prototyping.

OBSERVATIONS

- *Game developing companies expect their tools to allow easy prototyping and ability to design while implementing; large parts of the design is done with same tools as the actual development work.*
- Most common of the few mentioned issues were related to amount of bugs, or user interface design.
- Some organizations also considered that if they would have to increase the number of people in the project, they would probably have to adopt some sort of formalized project management system or more systematic development process model.
  - Teams are usually small enough so that formal work division tools are not that common.
- The case organizations did not apply self-made tools to a large degree. In fact, the case organizations were immigrating away from self-build tools towards third party systems.
  - From the perspective of game developers, this is a rather recent trend; before ~2010 all tools including engines were in-house systems.
  - On the other hand, large game studios such as Rovio or Supercell want people who can do *programming*, not people who can *use an engine X*.

BASICS OF GAME DESIGN

“If you cannot express your great idea with the maximum of 3 slides and 5 minutes, it is a bad idea.”

-GAME DESIGNER FROM ONE OF THE CASE COMPANIES, 2015

IN GENERAL

- Most game designs are based on a concept innovated by individuals.
  - No silver bullet, most design is coming up with a concept, making an abstraction of some activity out of it and making it absurd enough.
- Design and innovation are ad-hoc processes.
  - Usually team work, team pitching, coming up with a plan based on other media, trends, jokes, ideas…

EXAMPLE

- Cars are driving down the road in a crowded city center. → How about we make it an erratic elephant who tries to avoid damaging the city?
- Elephant is running down the road in a crowded city center. → We do not want to make it violent, so remove actual people and put in pigeons.
- Elephant is running down the pigeons in a crowded city center. → Make it cartoony so it is not animal abuse.
- Cartoon elephant is running down the pigeons in a crowded city center. → Where did the elephant come from? Maybe put clowns to chase it down?
- Cartoon elephant is running down the pigeons in a crowed city center while avoiding traps set by a bunch of clowns. → We need the weird factor, lets make the cartoon light posts alive so they try to trample the elephant?
- Cartoon elephant is running down the pigeons in a crowed city center while avoiding traps set by a bunch of clowns and avoiding being trampled by the light post-like stick figures. → Add story, maybe?.
- Cartoon elephant is running down the pigeons in a crowed city center while avoiding traps set by a bunch of clowns and avoiding being trampled by the light post-like stick figures in an open-world city landscape run crazy by a yodeling supervillain robbing the place with the pigeons and light post stick figures. The elephant is hunting them for **JUSTICE**. → **NAILED IT**

IN THE INDUSTRY

- Sources of innovation are mostly in existing game products and success stories.
  - However, today’s superhit is tomorrow's old news. We need to come up with new ideas constantly.
  - …And this being a university course the design-principle “go crazy” is absolutely encouraged.

- Start-ups are most business-driven in the game industry.
  - No large differences, but start-ups “do what pays”, instead of “own thing that makes money”.

FINDINGS FROM THE INDUSTRY

Game developers consider themselves to work in creative industry (in contrast of software ind.).

- However, when presented with the dilemma of money vs. fame, large majority selected money.

Game product design is driven by economic factors.

- Not art for art's sake; aim always in profitable product.
- Creativity and artistic merit secondary, but sought after attributes.

Technical design work relies on prototypes, which test out potential game concepts.

- Late changes allowed, even expected.

FINDINGS

The game products are designed with creative processes comparable to movies or any other artistic creation, but games are not intended to be art for art's sake, they are designed and intended to be commercial products which generate income.

In some organizations the most important design aspect was in developing "fun" product, but in the long run the organization was still aiming at commercial success.

OBSERVATIONS FROM THE INDUSTRY

	Case A	B	C	D	E	F	G
Objective of the design phase	Make something that sells, marketable in near future	Concept demo on technology, game mechanics	Test if the concept is fun	Good mechanics, game that sells	Test mechanics for concept, something that is fun.	Design of own thing, things selling are old six months	Design something we are very good at making
Design method	Idea pitching, prototypes, brainstorming	"Vision", group work	"Vision", Idea pitching, prototypes	"Vision", pen and paper	Brainstorming, prototypes, pen and paper	Prototyping, Vision	Vision
First vs. published	Major changes	Minor changes	Major changes	Major changes	Minor changes	Large major changes	Large technical, minor design
Level of details in the design	Functional prototype	Basic gameplay elements	Functional prototype	Core features, concept art	Basic gameplay elements	Core features, concept art	Basic gameplay elements
Effect of industry	Enforces features	Publisher sets requirements	Enforces upkeep, DLC	Changes to design	New customers, business models	Stabilizing effect on designs	"Marketing dictates success"
Most important designers	Producer	Lead designer, team	Producer	Lead designer	Team	Management	Lead designer
Innovation vs. money	Money first, then innovation	Money first, then innovation	Innovation, hopefully money	Money first, then innovation	Money first	Money first (free2play)	Money first, then innovation
Effect of marketing in design	"We design fun, management handles sales"	"Has to be profitable"	"Make fun demo and sell it"	"Business first"	"Business first"	"Good game sells"	"Finances has to be taken into account"
Sources of innovation	Movies, other games	Success stories, industry trends	Success stories	Prior experiences, old games	Platform possibilities, old games	Movies, books, TV, games, "portfolio of stuff"	Prior experiences, competition analysis

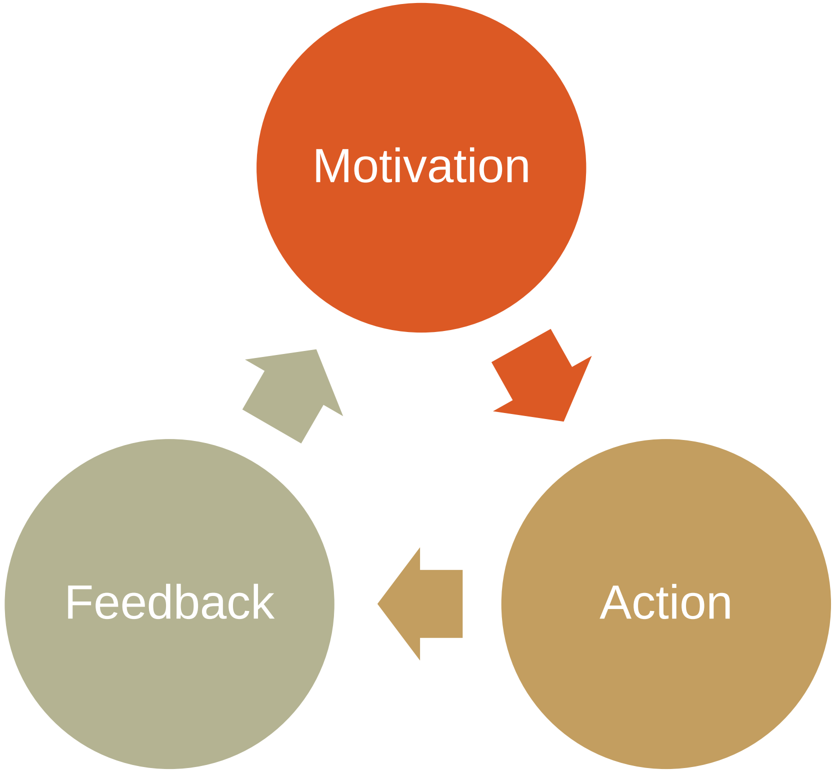
GAME DESIGN COMPONENTS

(WE WILL GET BACK TO SOME OF THIS WITH GAMIFICATION)

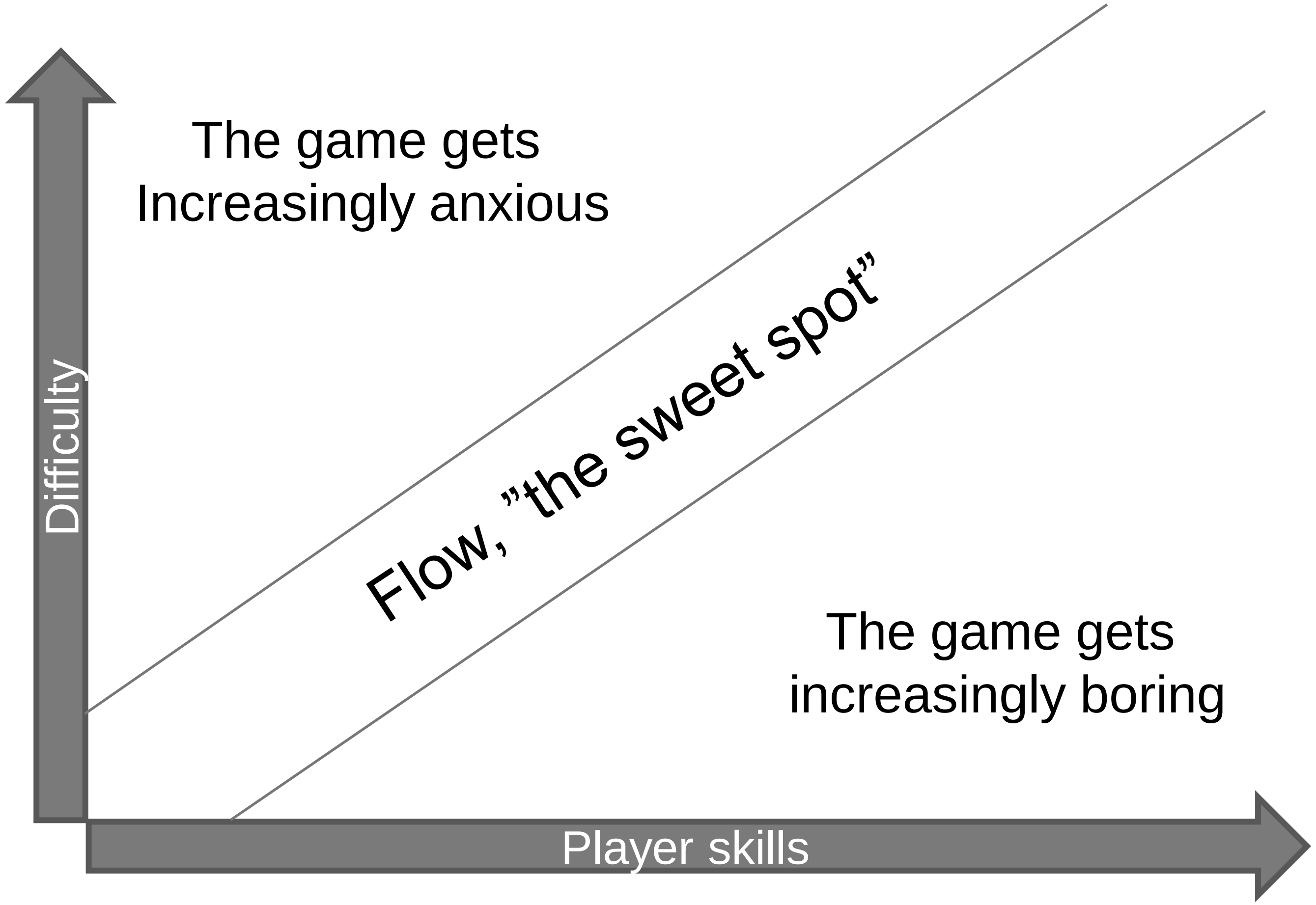
Mostly collected from Werbach & Hunter 2013 and Kapp 2012

Open your mind. LUT.
Lappeenranta University of Technology

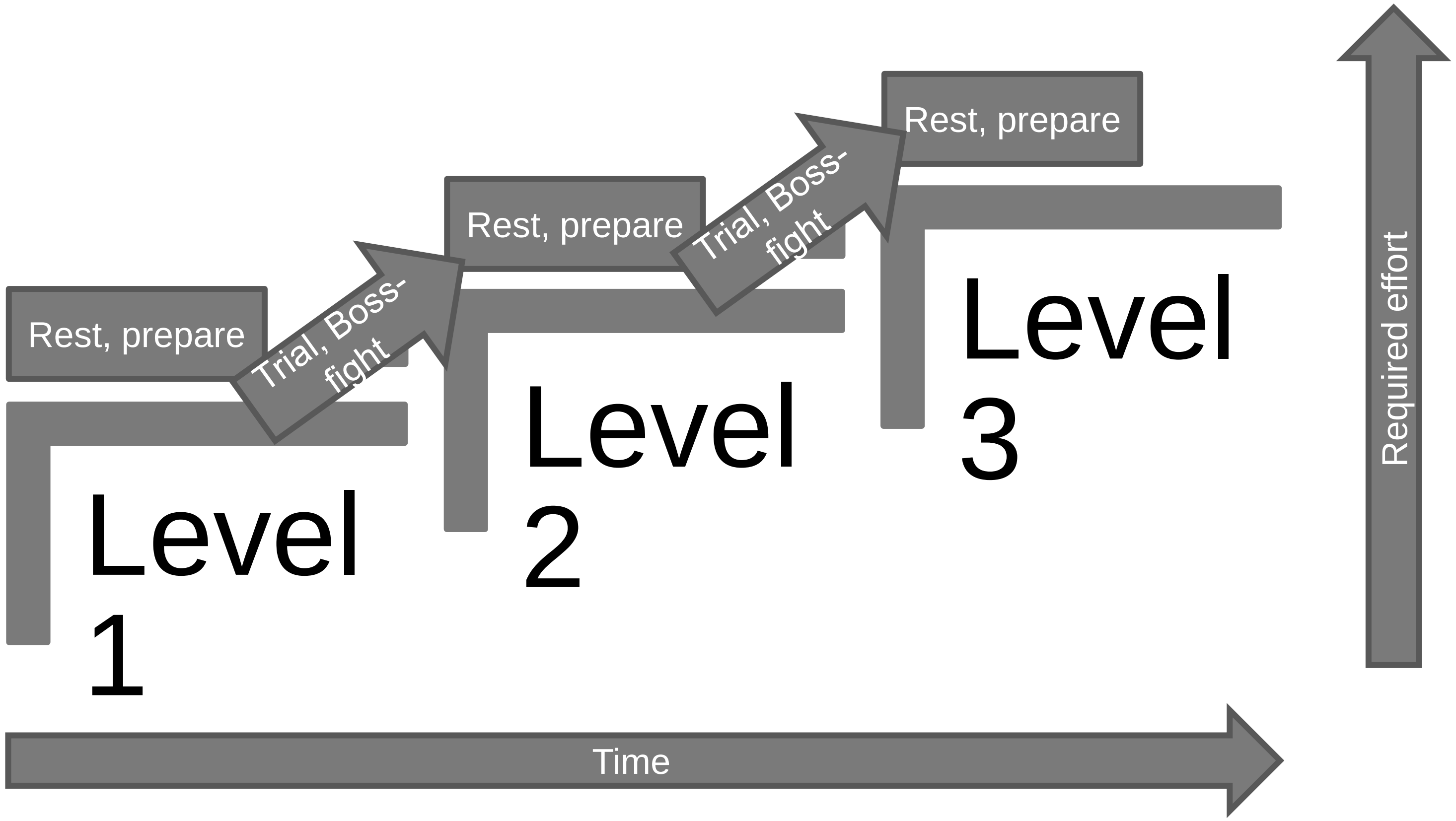
ACTIVITY CYCLE (FROM WERBACH & HUNTER)



”FLOW” BETWEEN BOREDOM AND ANXIENTY



PROGRESSION STAIRS



RE: GAME DESIGN DOCUMENT

The most important aspects written down:

- General theme, main characters.
- Genre (shooter, platformer, puzzle etc.)
- Mostly important features
- Story, if there is any.
- Technical decisions

- What are the things you aim for? In the course context, what are the mechanics that I should learn to implement this? What tutorials would actually benefit my design?

GAME DESIGN DOCUMENT: A TEMPLATE

See the template here:

https://docs.google.com/document/d/1zV330cE5tqbuQC0aJgMhEAn8nxzbb_UU0rOl0pCGJmU/edit?usp=sharing

Use this as your starting point!

CT60A5401 GAME DEVELOPMENT PROJECT

Comments, feedback, questions?

Email jussi.kasurinen@lut.fi